

ADRIAN SULICH

Contact details

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R^G www.researchgate.net/profile/Adrian_Sulich

Experimental skills

HR XRD

UV-VIS spectroscopy

Pulse radiolysis

Computer skills

MS Office ●●●●●

OriginLab ●●●●●

ISIS Draw ●●●●●

c++ language ●●●●●

Foreign languages

English ●●●●●

References

on request

Interdisciplinary researcher with experience in laboratory work, rich scientific achievements and science communication activity. The main research interests: physics, chemistry, and linguistics of the contemporary scientific language.

EDUCATION

- 28.05.2020 **PhD, Physics**, Institute of Physics PAS (IP PAS)
dissertation: „Single-crystal defect structure of selected high-symmetrical, multi-component oxides, containing rare earths”
supervisor: prof. dr hab. Wojciech Paszkowicz
assistant supervisor: dr Jarosław Z. Domagała
reviewers: dr hab. Zbigniew Kaszukur, dr hab. Grzegorz Kowalski
- 18.12.2019 **PhD, Chemistry**, Institute of Nuclear Chemistry and Technology (INCT)
dissertation: „Early radiolysis stages of the selected diluents and ligands for SANEX process”
supervisor: prof. dr hab. Krzysztof Bobrowski, dr. hab. Jan Grodkowski
reviewers: prof. dr hab. Andrzej Czerwiński, dr hab. inż. Mariusz Wójcik
- 09.07.2008 **MSc in Engineering, Chemical Technology**, Faculty of Chemistry, Warsaw University of Technology
field: Solid-State Technology and Characterization
- 23.01.2007 **PhD, Linguistics**, Faculty of Modern Languages, University of Warsaw
dissertation: “Metatekstual lexical units assigning the thematic structure of utterance”
supervisor: prof. dr hab. Bożenna Bojar
reviewers: prof. dr hab. Małgorzata Korytkowska, dr hab. Elżbieta Artowicz
- 28.06.2000 **MA, Polish Philology**, Faculty of Polish Studies, University of Warsaw

EMPLOYMENT

- 08.04.2020- Institute of Physics PAS, Laboratory of X-ray and Electron Microscopy Research, Research Assistant
- 2015- Institute of Computer Science PAS, multiple short-term contract jobs as Linguist in cooperation with Linguistic Engineering Group
- 08.10.2018- Institute of Physics PAS, Laboratory of X-ray and Electron Microscopy Research, Physicist
- 07.04.2020
- 01.08.2014- Institute of Physics PAS, Laboratory of Growth and Physics of Low Dimensional Crystals, Technician
- 31.08.2015
- 01.12.2008- Institute of Nuclear Chemistry and Technology, Centre for Radiation Research and Technology, Chemist
- 30.11.2013
- 2004-2008 Polish Scientific Publishers PWN, multiple short-term contract jobs as Linguist

SELECTED SCIENTIFIC ACHIEVEMENTS

- publications 23 articles (all disciplines)
1 monograph, co-authorship of 1 specialist dictionary
- presentations 2 contributed lectures and 9 posters at international conferences
2 seminars at foreign scientific institutions
1 oral report at the final meeting of the international scientific project

(full list of scientific achievements at the end of the CV)

ORGANIZATIONAL ACTIVITY

- 11-14.09.2018 The 10th International Workshop on Zinc Oxide and Other Oxide Semiconductors (IWZnO 2018), SGH Warsaw School of Economics, Warsaw, Poland, member of Local Organizing Committee

PARTICIPATION IN INTERNATIONAL SCIENTIFIC PROJECT

01.12.2008 ACSEPT – Actinide reCykling by SEParation and Transmutation
30.11.2013 FP7 Collaborative Project 211267, Institute of Nuclear Chemistry and Technology, investigator

INTERNATIONAL FELLOWSHIPS AND VISITS

04-08.09.2018 Institute for Single Crystals NASU (ISC NASU), Kharkiv, Ukraine
scientific visit in the frame of project: „Investigation of structure of $\text{Sr}_3\text{RE}_2(\text{BO}_3)_4$ (RE = Y, Gd) single crystals”
aims: discussion about the project, getting knowledge about single-crystals growth technology, developed at ISC NASU, giving a seminar about single-crystal defect structure characterization by means of HR XRD

13-16.11.2011 Commissariat à l'Energie Atomique, Marcoule, France, nuclear power plant and uranium concentration plant of AREVA company, Tricastin, France, professional visit
aim: getting knowledge about the French institutions connected with nuclear energy

01-27.03.2011 Chalmers University of Technology, Department of Chemical and Biological Engineering, Gothenburg, Sweden
scientific visit in the frame of ACSEPT project
aims: getting knowledge about the research done by foreign partner and presentation of the results of own research

03.-31.05.2008 Rostock Universität, Institute für Physikalische Chemie, Rostock, Germany
international student's exchange - Socrates-Erasmus Program
aim: realization of the part of study at a foreign university

NATURAL SCIENCES TEACHING EXPERIENCE

Institute of Physics PAS, 07.2018 (1 working day), 07.2017 (1 working day)
supervision of 2 students from the Warsaw University of Technology

Institute of Physics PAS, 10.2017 (1 working day)
working with a scientific intern from ISC NASU

SCIENCE COMMUNICATION

„Institute of Physics PAS Open Days” – editions: 2017, 2018

„Long Night of Museums in Warsaw” – editions: 2009 and 2011-2013
participation in presentations organized by Institute of Nuclear Chemistry and Technology

„Science Picnic of the Polish Radio and the Copernicus Science Center” – editions: 2009 2012
participation in presentations organized by Institute of Nuclear Chemistry and Technology

„Science Festival” – editions: 2010, 2011
participation in presentations organized by Institute of Nuclear Chemistry and Technology

Individual lecture in Maria Skłodowska Curie High School No. 23, Warsaw, with Institute of Nuclear Chemistry and Technology affiliation - 07.11.2012
„Discoveries by Maria Skłodowska-Curie in contemporary science and technique” (in Polish)

„The life and works of Maria Skłodowska-Curie. Women in science” – celebrations of 100 anniversary of Maria Skłodowska Nobel Prize awarding
participation in presentations organized by Institute of Nuclear Chemistry and Technology and Adam Mickiewicz University

SCIENTIFIC ACHIEVEMENTS

PART A)

NATURAL SCIENCES

ARTICLES IN JOURNALS FROM ISI MASTER JOURNAL LIST

- S. Mishra, W. Paszkowicz, **A. Sulich**, R. Jakiela, M. Ożga, and E. Guzewicz, *Electrical and structural properties of semipolar-ZnO/a-Al₂O₃ and polar-ZnO/c-Al₂O₃ films: A comparative study*, Materials, 2023, 16(1), 151. DOI: 10.3390/ma16010151.
- A. Wardak, D. M. Kochanowska, M. Kochański, M. Dopierała, **A. Sulich**, J. Gdański, A. Marciniak, and A. Mycielski, *Effect of doping and annealing on resistivity, mobility-lifetime product and detector response of (Cd,Mn)Te*, Journal of Alloys and Compounds, 2023, 936, 168280 (8pp.). DOI: 10.1016/j.jallcom.2022.168280.
- S. Mishra, B. S. Witkowski, R. Jakiela, Z. Khosravizadeh, W. Paszkowicz, **A. Sulich**, O. Volnianska, W. Woźniak and E. Guzewicz, *Cathodoluminescent imaging of ZnO:N films: Study of annealing processes leading to enhanced acceptor luminescence*, Physica Status Solidi A: Applications and Materials Science, 2022, 2200466 (11 pp.). DOI: 10.1002/pssa.202200466.
- M. Sarwar, B. S. Witkowski, **A. Sulich**, and E. Guzewicz, *Low-Temperature Cathodoluminescence of Nitrogen-Doped ZnO Films Deposited at Low-Temperature by Atomic Layer Deposition*, Acta Physica Polonica A, 2022, 141(2), 135-139. DOI: 10.12693/APhysPolA.141.135 (Proceedings of the 49th International School and Conference on the Physics of Semiconductors "Jaszowiec 2021").
- A. Sulich**, E. Łusakowska, W. Wołkanowicz, P. Dziawa, J. Sadowski, B. Taliashvili, T. Wojtowicz, T. Story, and J. Z. Domagala, *Unit cell distortion and surface morphology diversification in a SnTe/CdTe(001) topological crystalline insulator heterostructure: influence of defect azimuthal distribution*, Journal of Materials Chemistry C, 2022, 10(8), 3139-3152. DOI: 10.1039/d1tc05733b.
- S. Mishra, E. Przezdziecka, W. Wozniak, A. Adhikari, R. Jakiela, W. Paszkowicz, **A. Sulich**, M. Ozga, K. Kopalko, and E. Guzewicz, *Structural Properties of Thin ZnO Films Deposited by ALD under O-Rich and Zn-Rich Growth Conditions and Their Relationship with Electrical Parameters*, Materials, 2021, 14(14), 4048 (21pp). DOI: 10.3390/ma14144048.
- E. Przezdziecka, E. Guzewicz, D. Jarosz, D. Snigurenko, **A. Sulich**, P. Sybilski, R. Jakiela, W. Paszkowicz, *Influence of oxygen-rich and zinc-rich conditions on donor and acceptor states and conductivity mechanism of ZnO films grown by ALD – experimental studies*, Journal of Applied Physics, 2020, 127, 075104 (14pp).
- A. Sulich**, J.Z. Domagala, J. Härtwig, A. Shekhovtsov, M. Kosmyna, L.V. Gudzenko, and W. Paszkowicz, *Nature and Spatial Distribution of Extended Defects in Czochralski-Grown Ca₃RE₂(BO₃)₄ (RE = Y, Gd) Orthoborate Single Crystals*, J. Phys. D: Appl. Phys., 2019, 52, 055102 (13pp).
- B.I. Lazoryak, D.V. Deyneko, S.M. Aksenov, V.V. Grebenev, S.Yu. Stefanovich, K.N. Belikov, M.B. Kosmyna, A.N. Shekhovtsov, **A. Sulich**, W. Paszkowicz, *Influence of lithium and magnesium on the real structure and dielectric properties of Ca₉Y(VO₄)₇ single crystals*, CrystEngComm, 2018, 20, 6310–6318.

L.V. Gudzenko, M.B. Kosmyna, A.N. Shekhovtsov, W. Paszkowicz, **A. Sulich**, J.Z. Domagała, P.A. Popov, S.A. Skrobov, *Crystal growth and glass-like thermal conductivity of $\text{Ca}_3\text{RE}_2(\text{BO}_3)_4$ ($\text{RE} = \text{Y, Gd, Nd}$) single crystals*, Crystals 7(3):88, March 2017, DOI: 10.3390/cryst7030088.

A. Sulich, J. Grodkowski, J. Mirkowski, R. Kocia, *Reactions of ligands from BT(B)P family with solvated electrons and benzophenone ketyl radicals in n-octanol solutions. Pulse radiolysis study*, Journal of Radioanalytical and Nuclear Chemistry, Vol. 299, number 3 (2014).

ANNUAL REPORTS OF INCT

J. Grodkowski, R. Kocia, J. Mirkowski, M. Nyga, **A. Sulich**, T. Szreder, *Halide radicals in ionic liquids. Pulse radiolysis study*, Annual Report 2012, INCT, 17-18.

J. Grodkowski, R. Kocia, J. Mirkowski, M. Nyga, **A. Sulich**, T. Szreder, *Oxidizing radicals and their reactivity in ionic liquid based on NTf_2^- anion*, Annual Report 2011, INCT, 19-20.

J. Grodkowski, R. Kocia, J. Mirkowski, M. Nyga, **A. Sulich**, A. Skrzypczak, T. Szreder, *Radiation degradation of some ionic liquids with methyltributylammonium cation, MeBu_3N^+* , Annual Report 2010, INCT, 21-23.

REPORT AT THE FINAL MEETING OF INTERNATIONAL SCIENTIFIC PROJECT

ACSEPT Meeting, Montpellier, France, 08.09.2012,
oral report: „Assessment and Studies on Improving Radiation Stability of Ligands”,
J. Grodkowski, **A. Sulich**
(I presented the report as a substitute for professor Grodkowski, who was not able to arrive in France due to a serious disease).

PRESENTATIONS AT INTERNATIONAL CONFERENCES

Contributed lectures

E-MRS Fall Meeting 2021, online, 20-23.09.2021,
“Surface Nanoripples Formation in $\text{SnTe}(001)/\text{CdTe}(001)/\text{GaAs}(001)$ Topological Crystalline Insulator Heterostructure”,
A. Sulich, E. Łusakowska, W. Wołkanowicz, P. Dziawa, T. Story, J.Z. Domagała.

ATALANTE 2012 – Nuclear Chemistry for Sustainable Fuel Cycles, Montpellier, France, 02-07.09.2012,
„Research on Radiation Chemistry of Selected Substances Applied in SANEX Process”,
A. Sulich, J. Grodkowski, J. Mirkowski, R. Kocia, M. R. StJ. Foreman, M. J. Hudson.

Poster presentations

Topological Matter Conference 2023, Athens, Greece, 28-31.03.2023,
“Can a $\text{SnTe}/\text{CdTe}(001)$ heterostructure be a promising material to open an energy gap in its topological surface states?”,
A. Sulich, E. Łusakowska, W. Wołkanowicz, P. Dziawa, J. Sadowski, B. Taliashvili, T. Wojtowicz, T. Story, and J. Z. Domagała.

- 50th International School & Conference on the Physics of Semiconductors "Jaszowiec 2022", Szczyrk, Poland, 4-10.06.2022,
 "SnTe Unit Cell Distortion and Surface Morphology Diversification in (001)-oriented Heterostructure: Influence of Defect Azimuthal Distribution",
A. Sulich, E. Łusakowska, W. Wołkanowicz, P. Dziawa, J. Sadowski, B. Taliashvili, T. Wojtowicz, T. Story, and J. Z. Domagala.
- The 2018 Europhysical Conference on Defects in Insulating Materials, University of Bydgoszcz, Bydgoszcz, Poland, 08-13.07.2018,
 "High-resolution XRD study on selected Czochralski-grown rare-earth containing borates and gallates",
A. Sulich, J. Z. Domagala, W. Paszkowicz, M. Berkowski, A. Shekhovtsov, M. Kosmyna.
- 2017 E-MRS Fall Meeting, Warsaw, Poland, 18-21.09.2017,
 Defect Structure Characterization of Czochralski-Grown $\text{Ca}_3\text{RE}_2(\text{BO}_3)_4$ (RE = Y, Gd) Single Crystals",
A. Sulich, J.Z. Domagala, J. Härtwig, W. Paszkowicz, A. Shekhovtsov and M. Kosmyna.
- 46th "Jaszowiec" International School & Conference on the Physics of Semiconductors, Szczyrk, Poland, 17-23.06.2017,
 "Structural Anisotropy of MBE-grown CdTe/SnTe/CdTe//GaAs(001) Films",
A. Sulich, E. Łusakowska, P. Dziawa, J.Z. Domagala.
- The 13th International School and Symposium on Synchrotron Radiation in Natural Science (ISSRNS'2016), Ustroń-Jaszowiec, Poland, 13-18.06.2016,
 „Crystal lattice deformation in (001)-oriented CdTe/SnTe/CdTe trilayers MBE-grown on 2° offcut GaAs”,
A. Sulich, J.Z. Domagala, E. Łusakowska, P. Dziawa.
- 14th International Congress of Radiation Research, Warsaw, 28.08-01.09.2011,
 „Studies on Improving Radiation Stability of Substances Applied In the Recycling of a Spent Nuclear Fuel”,
A. Sulich, J. Grodkowski, J. Mirkowski, R. Kocia, M. Foreman.
- 5th European Young Investigators Conference (EYIC), Słubice (Poland) and Frankfurt/Oder (Germany), 22-26.06.2011,
 „Benzophenone as a probe in the pulse radiolysis of cyclohexanone and 1-octanol, diluents for procedures in minor actinides extraction”,
A. Sulich, J. Grodkowski, J. Mirkowski, R. Kocia, M. Foreman.
- 4th European Young Investigators Conference (EYIC), Słubice, Poland, 18-21.06.2009,
 „Pulse radiolysis of CyMe4-BTP and CyMe4-BTBP in octanol”,
A. Sulich, J. Grodkowski, J. Mirkowski, R. Kocia, M. Foreman.

SEMINARS AT FOREIGN SCIENTIFIC INSTITUTIONS

- Institute for Single Crystals NASU, Kharkov, Ukraine, 05.09.2018,
 "Characterization of Extended Defects and Their Spatial Distribution in Single Crystals Using the High Resolution X-ray Diffraction: Application to $\text{Ca}_3\text{Y}_2(\text{BO}_3)_4$ Orthoborate",
A. Sulich.
- Department of Chemical and Biological Engineering, Chalmers University of Technology, Gothenburg, Sweden, 11.03.2011,
 „Institute of Nuclear Chemistry and Technology (INCT) - a short presentation about us”,
A. Sulich.

PART B)

HUMAN SCIENCES

INTERDISCIPLINARY PUBLICATIONS – LINGUISTICS AND SEMIOTICS OF SCIENTIFIC LANGUAGES

- A. Sulich, *Toward the Language of Reciprocal Space Maps* (in Polish), *Humanistyka i Przyrodoznawstwo*, 2018, 24, 519-562. A. Sulich *How the chemists talk about their research?* (in Polish), *Acta Philologica* vol. 46, (2015), s. 103-108.
- A. Sulich, *Toward the morphological structure of chemical compounds names – by the example of phrase: kwas 4,4'-[1-[(5-[(4'-cyjanobifenylo-4-ylo)oksy]pentylo)oksy)karbonylo]etyleno}dibenzoikowy* (in Polish), *Acta Philologica*, 44, 2013, s.153-160.
- A. Sulich, *Naming of chemical elements in various languages in the world* (in Polish), *Acta Philologica*, 42, 2012, s.5-13.
- A. Sulich, *How the names of chemical elements represent knowledge* (in Polish), *Studia Semiotyczne*, vol. XXVII, jubilee edition to celebrate the 40th anniversary of the Polish Semiotic Society, 2010, s.93-133.
- A. Sulich, *How the names of chemical elements represent knowledge* – translation of the previous article to English by the Polish Semiotic Society at 2013, published in *Studia Semiotyczne – English Supplement*.
- A. Sulich, *Two scientific worlds, two genres? – the structure and the functioning of academic articles in the fields of chemistry and linguistics* (in Polish), *Acta Philologica*, 36, 2009, s.13-22.

POLISH-STUDY PUBLICATIONS

Monograph published by a publisher from Polish Ministry of Science and Higher Education List

- A. Sulich, *Polish thematizers. Lexical units establishing thematic structure of a discourse* (in Polish) Warsaw 2008, Semper Publisher

Articles and chapters in monographs

- A. Sulich, *Stereotype of 'Hungarian uprising in 1956 year' in the texts from «Gazeta Wyborcza» and «Gazeta Polska»* (in Polish), [in:] Budapest in 1956. Uprising of the hope (in Polish) ed. B. Bojar, Warsaw 2016, 131-148.
- A. Sulich, *How many lexical units of «co do_» form do exist?* (in Polish), *Acta Philologica*, nr 40, 2011, 11-22.
- A. Sulich, *Evaluation of statements in respect of their coherence with the Grice's maxim of relevance* (in Polish), [in:] *Antinomies of values* (in Polish), ed. A. Oskiera, Łódź 2007, s.265-274.
- A. Sulich, *«Jeśli chodzi o x, (to) q» as a thematizer referring to a previous statement* (in Polish) [in:] *Studies on the contemporary Polish. Grammar, semantics, pragmatics* (in Polish), ed. A. Dobaczewski, Toruń 2007, 185-197.
- A. Sulich, *Semantic structure of statements with the predicate «chodzi o...»* (in Polish) [in:] *Language, Literature, Didactics* (in Polish), ed. J. Opoka and A. Oskiera, Łódź 2003, vol. I, 121-128.

A. Sulich, *Linguistic image of 'father' in the contemporary Polish language* (in Polish), *Poradnik Językowy*, 3, 2002, s. 3-20.

Dictionary

B. Alberski, J. Andrejewicz, U. Andrejewicz, A. Andrzejczuk, P. Batko, M. Brodzińska, H. Bukowiecka, L. Drabik, J. Filipczak, A. Grzeszak, E. Hajnicz, B. Itoya, E. Kaczmarek, M. Kalużna-Gołąb, N. Kocyba, M. Kozłowska, B. Linsztet, A. Łodzińska, M. Maciejewska, A. Norwa, M. Opacki, A. Patejuk, A. Przepiórkowski, J. Rąbiega-Wiśniewska, P. Rosalska, N. Skubida, F. Skwarski, A. Stankiewicz, **A. Sulich**, M. Szczyszek, J. Szymczak, M. Świdziński, L. Wiśniakowska, M. Woliński, A. Wójcicka, A. Zagajewska, M. Zawisławska, M. Zgondek, N. Żochowska, S. Żurowski, 2018, *Walenty* (2018-06-29), CLARIN-PL digital repository, <http://hdl.handle.net/11321/586>.

CONFERENCE PRESENTATIONS

Interdisciplinary presentations – linguistics and semiotics of scientific languages

National Semiotic Symposium (in Polish), Warsaw, Warsaw University of Life Sciences, 9.12.2016 r,
contributed lecture: „The semantics of Reciprocal Space Maps”,
A. Sulich.

„Coherence of a Specialist Text” (in Polish), Warsaw, Institute of Anthropocentric Linguistics and Cultural Studies, University of Warsaw, 19.09.2014,
contributed lecture: „Abbreviated forms of ionic liquids names – are they normal acronyms or chemical formulas?” (in Polish)
A. Sulich.

Polish-study presentations

„Antinomies of values” (in Polish), Smardzewice, Poland, 19-21.05.2004,
contributed lecture: „Evaluation of statements in respect of their coherence with the Grice’s maxim of relevance” (in Polish),
A. Sulich.

„Language – Literature – Didactics” (in Polish), Poronin, Poland, 25-27.04.2002,
contributed lecture: „Semantic structure of statements with the predicate *chodzi o...*” (in Polish)
A. Sulich.

SEMINARS AT SCIENTIFIC INSTITUTIONS – LINGUISTICS AND SEMIOTICS OF SCIENTIFIC LANGUAGES

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 19.02.2018 (part 1) and 26.02.2018 (part 2),
„The language of chemical equations” (in Polish)
A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 20.02.2017 (part 1) and 27.02.2017 (part 2),
„‘Crystal’ as a mental object in specialist texts in the field of solid state physics and related sciences” (in Polish)
A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 07.12.2015 (part 1) and 14.12.2015 (part 2),

„On one occasion... there was a conversation between a human and an X-ray diffractometer, in other words: Diffraction Curve Imaging as a kind of text” (in Polish),

A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 02.12.2013 (part 1) and 09.12.2013 (part 2),

“Abbreviated forms of ionic liquids names – are they normal acronyms or chemical formulas?” (in Polish)

A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 20.01.2013,

“How the chemists talk about their research?” (in Polish),

A. Sulich.

Faculty of Philosophy and Sociology, University of Warsaw, 21.12.2012,

„The semiotics of chemical-compounds-structural-elements classification systems in the contemporary chemistry” (in Polish),

lecture in the frame of “Sign - Language - Reality” Project (in Polish), carried out by the Polish Semiotic Society,

A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 28.04.2010,

„The literature concerning specialist languages of chemistry. An attempt of overview” (in Polish)

A. Sulich

Faculty of Philosophy and Sociology, University of Warsaw, 15.01.2010,

„The periodic table of elements as a model of the knowledge ordering” (in Polish),

lecture in the frame of “Sign - Language - Reality” Project (in Polish), carried out by the Polish Semiotic Society,

A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 20.01.2009 (part 1) and 17.02.2009 (part 2),

„The languages of chemistry. An attempt of a classification” (in Polish),

A. Sulich.

The Faculty of Modern Languages, University of Warsaw, Department of Hungarian Studies, 06.01.2009 (part 1) and 13.01.2009 (part 2),

„The periodic table of elements as a model of the knowledge ordering” (in Polish),

A. Sulich.

Faculty of Philosophy and Sociology, University of Warsaw, 18.01.2008,

„How the names of chemical elements represent knowledge” (in Polish)

lecture in the frame of “Sign – Language - Reality” Project (in Polish), carried out by the Polish Semiotic Society,

A. Sulich.